

Heart Disease Severity Prediction & Diagnostic Screening

A Clean Object-Oriented Machine Learning Pipeline with In-Depth Exploratory Data Analysis

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1. Executive Summary & Objectives

Coronary artery disease remains the leading cause of mortality worldwide. The objective of this project was to design, implement, and validate an end-to-end Machine Learning pipeline to predict disease severity and clinical presence using the multicenter UCI Heart Disease cohort. The project emphasizes **rigorous exploratory data analysis (EDA)**, **data leakage elimination** via scikit-learn ColumnTransformers, and a clean, maintainable **Object-Oriented Programming (OOP)** architecture.

2. Dataset & Multicenter Clinical Cohort

The UCI Heart Disease dataset aggregates **920 patient records** across four geographic clinical centers: Cleveland Clinic (USA), Hungarian Institute of Cardiology (Hungary), University Hospital Zurich (Switzerland), and V.A. Medical Center Long Beach (USA). The dataset contains 16 features encompassing patient demographics, clinical symptoms, resting vitals, electrocardiographic (ECG) metrics, and exercise stress test parameters.

3. Key Insights from Exploratory Data Analysis (EDA)

Prior to modeling, a thorough visual and statistical EDA led to the following critical data decisions:

- **Handling Extreme Missingness:** Features *ca* (66.4% missing), *thal* (52.8% missing), and *slope* (33.6% missing) were removed because cross-center protocol discrepancies caused systematic data absence. Imputing over 50% of missing values would introduce heavy bias.
- **Biological Anomaly Correction:** 172 records in the Hungarian and Swiss subsets contained *chol* = 0 mg/dl and 1 record had *trestbps* = 0 mm Hg (physiologically impossible). These invalid zeroes were converted to NaN and imputed using median/MICE methods rather than discarding 19% of the cohort.
- **Chest Pain Paradox:** Over 78% of patients presenting with *asymptomatic chest pain* tested positive for coronary disease due to referral bias. Categorical variables (*cp*, *dataset*, *restecg*) were encoded using One-Hot Encoding (drop first).
- **Correlation Drivers:** Exercise-induced ST depression (*oldpeak*, $r = +0.43$) and maximum heart rate achieved (*thalch*, $r = -0.39$) emerged as the strongest physiological discriminators.

4. Object-Oriented Pipeline Architecture & Leakage Prevention

The codebase was refactored from a monolithic script into five decoupled, single-responsibility classes: **DataLoader** (raw ingestion & anomaly replacement), **DataPreprocessor** (ColumnTransformer encapsulation), **ModelTrainer** (Stratified 5-fold GridSearchCV & pipeline assembly), **ModelEvaluator** (multiclass & binary metrics), and **HeartDiseasePipeline** (high-level orchestrator).

Data Leakage Elimination: In standard scripts, fitting imputers or scalers globally on the dataset before splitting causes test data information to leak into training folds. Here, preprocessing is strictly wrapped inside a Pipeline, ensuring statistics are computed exclusively on training folds during cross-validation.

5. Experimental Results & Performance Benchmarks

The pipeline was evaluated on an 80/20 Stratified train/test split. We benchmarked a tuned **XGBoost Classifier** against a **Random Forest Baseline** across both the 5-class severity task and the standard binary clinical screening task (diameter narrowing > 50%).

Classification Task	Model Architecture	Test Accuracy	Macro F1	Weighted F1
5-Class Severity (0-4)	Random Forest (Baseline)	57.61%	0.4454	0.6035
5-Class Severity (0-4)	Tuned XGBoost Pipeline	58.70%	0.3521	0.5719
Binary Screening (0 vs 1+)	Binary XGBoost Pipeline	83.15%	0.8290	0.8310

6. Analysis of Class Imbalance & Clinical Implications

A key finding from this research is the pronounced impact of target class imbalance in multiclass staging. Class 4 (severe disease) constitutes only **3.0% (28 samples)** of the entire dataset, compared to **44.7% (411 samples)** in Class 0. While 5-class staging achieves ~59% accuracy due to sparse minority data, the model achieves **83.15% accuracy** on the binary screening task, confirming that the learned feature representations reliably identify cardiac risk.

7. Summary of Deliverables & Conclusion

This project successfully delivers an end-to-end, scientifically validated machine learning pipeline. Deliverables include:

- Clean OOP Python Engine:** heart_disease_prediction.py with full pipeline orchestration.
- Visual EDA Suite:** 9 high-resolution charts in eda_charts/ and an interpreted EDA_Report.md.
- Interactive Notebook:** heart_disease_prediction.ipynb for step-by-step exploration.
- Exported Production Artifact:** Serialized xgboost_model.pkl ready for API serving.